AUTOMATED VEHICLE ACCESS CONTROL SYSTEM

FPGA-Based FSM Controller Proposal

Course: FPGA/HDL Logic Design

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Table of Contents

1. Introduction

Background

Current barrier systems often rely on microcontroller-based interrupts, which can introduce latency leading to potential mechanical damage during emergency obstruction events.

Objectives

To design a deterministic, hardware-level Finite State Machine (FSM) implemented in Verilog. By bypassing software-based interrupt layers, we aim to achieve sub-millisecond response times for safety-critical interlocks.

Scope

The project covers the development of a modular RTL design governing barrier motor logic, including UART communication, state transitions, and PWM positioning.

2. Proposed Solution & Architecture

We propose a modular RTL (Register Transfer Level) design to govern the barrier motor. The system architecture decouples the communication interface (UART) from the safety-critical state transition logic.

This design ensures a "fail-safe" operation where safety logic (obstruction detection) is hardcoded at the highest priority within the FSM to ensure the gate reverses direction immediately upon detection.

3. Key Features & Functionality

Deterministic FSM Logic
The barrier operates through distinct states (IDLE, OPENING, OPEN, CLOSING) with hardcoded priority for emergency overrides.

UART-Based Authorization
A hardware-native UART receiver module processes serial payment data, ensuring real-time authorization without external library dependencies.

PWM Motor Positioning
Utilizes a custom PWM generator core to translate state signals into precise motor/servo movement commands ($0°$ to $90°$).

Hardware-Level Safety Interlocking
The obstruction sensor signal is treated as an asynchronous input, ensuring the gate reverses within a single clock cycle if an object is detected.

4. Required Components & IP Cores

- **Vivado Design Suite:** For synthesis, timing analysis, and simulation.
- **FSM_Controller:** Core logic governing barrier behavior.
- **UART_RX_Module:** Serial decoder for "Payment Received" signals.
- **PWM_Generator:** Precision motor position controller.
- **Input_Synchronizer:** Prevents metastability on asynchronous sensor inputs.

5. Implementation Timeline

Phase	Milestone	Duration
Phase 1	Requirement Analysis & FSM State Diagram	Week 1
Phase 2	RTL Development (UART, FSM, PWM)	Week 2
Phase 3	Behavioral Simulation & Edge Case Testing	Week 3
Phase 4	Final Documentation & Presentation	Week 4

6. References (IEEE Format)

[1] L. Dilshan, "UART using Verilog," GitHub, 2026. [Online]. Available: https://github.com/LasiduDilshan/UART-using-Verilog

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[3] Devipriya1921, "Traffic Light Controller using Verilog," GitHub, 2026. [Online]. Available: https://github.com/Devipriya1921/Traffic-Light-Controller-using-Verilog